Problem Challenge 3 (medium)

Problem Statement

You are given a 2D matrix containing different characters, you need to find if there exists any cycle consisting of the same character in the matrix.

A **cycle** is a path in the matrix that starts and ends at the same cell and has **four or more cells**. From a given cell, you can move to one of the cells adjacent to it - in one of the four directions (up, down, left, or right), if it has the **same character value** of the current cell.

Write a function to find if the matrix has a cycle.

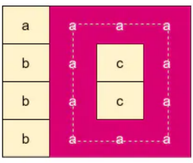
Example 1

Input: matrix =

a	a	a	a
b	a	c	a
b	a	c	a
b	a	a	a

Output: true

Explanation: The given matrix has a cycle as shown below:



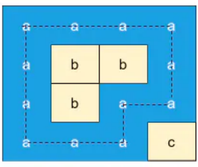
Example 2

Input: matrix =

a	a	a	a
a	b	b	a
a	b	a	a
a	a	a	c

Output: true

Explanation: The given matrix has one cycle as shown below:



Example 3

Input: matrix =

Output: false

Explanation: The given matrix has no cycle.

a	b	e	b
b	b	c	b
b	c	c	d
d	c	d	d

Try it yourself

Try solving this question here:

```java
import java.util.*;
class Solution {

  public boolean hasCycle(char[][] matrix) {
    // TODO: Write your code here
    return false;
  }
}
```

Test Case | Run Code Result | Stdout

[["a", "a", "a", "a"], ["b", "a", "c", "a"], ["b", "a", "c", "a"], ["b", "a", "a", "a"]]

Use Example Testcases | Run | Submit